Journal of Building Engineering 75 (2023) 106875

Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe

The role of socio-demographic and economic characteristics on energy-related occupant behavior

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ARTICLE INFO

Keywords:
Buildings
Energy consumption behavior of occupants
Hotel buildings
Energy saving strategies
Changing occupants' behavior
Education-based interventions
Incentives

ABSTRACT

Building-related sectors use more energy than all other energy consuming sectors. Energy consumption behavior of occupants and their socio-demographic profiles are some of the key factors affecting building energy consumption. This study aims to understand the relationships between the socio-demographic and economic characteristics of occupants and their energy-related behavior. To achieve this aim, six hypotheses are developed and tested over three steps: (1) identification of the changes in the occupants' energy-related behavior before and after being exposed to energy-saving interventions, (2) measurement of the socio-demographic profiles of the occupants, and (3) use of several machine learning methods to capture the relationships and test the hypotheses. Using decision tree learning to interpret the results, we find that education level, income, and age have the largest impact to predict the energy consumption behavior of occupants. The results also show four occupant profiles prone to switch to lower energy use: (1) age between 20 and 39 years old, education level of high school degree or lower, and income below 20 k; (2) age of 30 years old or younger, education level of high school degree or lower, and income above 100 k; (3) age of 40 years old or older, education level of bachelor's degree or lower, and income below 20 k; and (4) age of 59 years old or younger, education level of Master's degree or higher, and income above 100 k. This study can help decision-makers (e.g., utility companies, policy makers) to tailor incentive programs and policies to ultimately lower global building energy consumption.

1. Introduction

Building-related sectors use more energy than all other energy-consuming sectors [1]. They are responsible for 30% of the world's total energy consumption and 27% of total energy-related CO_2 emissions. In 2021, building-related sectors were also estimated to consume about 35% of global electricity [2]. In the United States (U.S.), the building sector was responsible for about 29% of total energy consumption and about 40% of total electricity consumption in 2022 [3]. In 2021, after restrictions related to the COVID19 pandemic were lifted, energy demand in the building sector increased by nearly 4% compared to 2020, the largest annual increase recorded in the last decade [2]. Moreover, the U.S. Energy Information Administration [4] forecasted a 50% increase in global energy consumption by 2050 compared to 2020 due to rapid change in socio-demographic and economic characteristics.

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https://doi.org/10.1016/j.jobe.2023.106875
Received 21 December 2022; Received in revised form 10 May 2023; Accepted 16 May 2023
Available online 31 May 2023
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The energy consumption behaviors of occupants—actions or reactions of a person to adapt to their ambient environmental conditions such as temperature, indoor air quality and sunlight—is one of the key factors affecting building energy consumption [5–17]. Many studies have investigated the relationship between the socio-demographic and economic characteristics of occupants (e.g., age, gender, education level, ethnic background, income, and employment status) and their energy consumption behavior in buildings [6, 18–32]. These studies have highlighted the importance of having records of the socio-demographic and economic characteristics of occupants inside different building types (e.g., residential, and commercial) for monitoring and controlling the energy consumption behaviors of occupants, and accordingly lower buildings' energy usage. Studies have also illustrated that the relationship between socio-demographics and energy consumption behavior of occupants is complex and further needs to be investigated [6,20,23,27,29, 31–33]. Many studies have predicted that socio-demographic variables such as age, gender, education level, ethnic background, income, and employment status have impact on occupants' energy-related behavior [29,30,34–38] while other studies found that socio-demographic variables have little to no impact on occupants' energy-related behavior [18,21,31,35,39–41]. There are several reasons why these studies reported different findings. One reason is the use of different factors or methodologies. For example, a study can focus solely on the impact of age, while another study considers numerous factors such as age, education, and income. Additionally, the relationship can also be influenced by a variety of external factors, such as climate, building age and construction, and the availability of energy-efficient technologies. Another factor is the use of different metrics to measure energy consumption. Some studies focus solely on electricity consumption, while others can also consider natural gas, oil, or other energy sources. Despite their contribution, these studies show mixed, unclear, and/or inconsistent results. Additionally, these studies have not investigated and analyzed which socio-demographic and economic characteristics of occupants are more likely to change their energy consumption behavior when exposed to interventions and incentives. To address this knowledge gap, the main contributions of this study are: (1) the identification of both the most effective strategies to change occupants' energy consumption behavior by using education-based interventions and incentives, and the socio-demographic and economic variables that are prone to change their energy consumption behavior; and (2) the measurement of occupants' energy consumption behavior in hotel buildings which have standardized or uniform structures and provide the same level accessibility and controllability to building features for all occupants. This standardized environment in hotel buildings offers an equal opportunity for all occupants to accurately measure their energy use patterns' relationship with their socio-demographic and economic characteristics as opposed to previous studies that have been conducted mainly in residential buildings such that the nature of residential buildings (e.g., layout and insulation type) can differ significantly based on the socio-demographic characteristics of their occupants (e.g., income, employment status).

In short, this study aims to better understand the relationships between energy-use behavior of occupants and their socio-demographic and economic characteristics for reducing building energy consumption. More specifically, this study highlights the importance of several socio-demographic and economic variables such as age, gender, education level, ethnic background, income, and employment status on occupants' energy-use behavior when exposed to education-based interventions.

To achieve this aim, a set of six hypotheses is developed to examine the relationship between the socio-demographic and economic characteristics of occupants and their energy-related behavior (i.e., energy-use behavior and energy-use profiles). Then, changes in occupants' energy-related behavior are identified before and after the occupants were exposed to proposed interventions and incentives by collecting data from a longitudinal study of two surveys conducted by Refs. [13,14]. Next, socio-demographic and economic profiles of the occupants are measured and recorded. After that, the relationship between the profiles of the occupants and their energy-related behavior are studied using traditional statistical methods as well as machine learning methods (e.g., linear regression, decision tree learning, support vector machines) that can better capture complex relationships as is often in the case with behavior [42, 43]. Finally, the study hypotheses are tested.

The findings of this study can help decision-makers (e.g., utility companies, policy makers) propose targeted strategies based on occupants' socio-demographic and economic profile to lower energy consumption in buildings. The next section offers a literature review and a bibliometric analysis.

2. Background

2.1. Literature review

Many studies have studied the relationship between the socio-demographic and economic characteristics of occupants (age, gender, education level, ethnic background, income, and employment status) and their energy consumption behavior to reduce building energy consumption as shown in Table 1. Although the studies listed in Table 1 examined various building types, including residential (single and multi-family housing), offices, and dormitories, the results were inconclusive regarding the relationship

Table 1
Literature Review about Socio-demographic and economic Characteristics of Occupants and their Energy Consumption Behavior.

Socio-Demographic and Economic Characteristics	Positive Relationship	Negative Relationship	Mixed Relationship	No Relationship
Age	[34–36,44]	[23,38,40,45,46]	[18,26,29,47]	[31,37]
Gender	[29,36,48–50]		[35,51]	[52,53]
Education Level	[54]	[21,31,55,56]	[40,57]	[29]
Race/Ethnicity	[25,56]		[21]	[39]
Income	[18,24,26,32,41,56,58,59]	[60]	[31]	[61]
Employment Status	[27,46,62]		[22]	[31,41]

Here, the literature is reviewed, considering each characteristic individually.

between the socio-demographic and economic characteristics of occupants and their energy-related behavior regardless of the building type.

Age. Few studies found that that there is a positive relationship between age and energy-relate behavior—as the age of the individuals increase, their energy consumption increase as well [34–36,44]. Another set of studies found a negative relationship between age and energy-relate behavior [23,38,40,45,46]. Few studies have found a mixed or no relationship [18,26,29,31,37,47].

Gender. Several studies found that females are more likely to adopt energy saving practices than males [29,48–50]. Other studies stated that men consume less energy compared to women [36]. Only a limited number of studies found that there is mixed relationship between gender and energy consumption behavior of occupants in buildings [35,51].

Education Level. Few studies found that there is positive relationship between education level and energy saving actions—as the education level of the occupants increase, their energy consumption increase as well [54]. Many studies determined that there is negative relationship between education level and energy saving actions—as the occupants education level increase, their energy consumption decrease [21,31,55,56]. Few studies also found that mixed to no impact between occupants' education level on energy consumption of buildings [29,40,57].

Race/Ethnicity. [56] found that Whites consume less energy than Blacks or African American and other minority groups. However [25], demonstrated that in residential buildings, non-Hispanics and/or Whites consume more energy compared to Hispanics and/or non-Whites. Yet, other studies found mixed or no relationship with ethnicity and race [21,39].

Income [60]. indicated that low-income occupants are less able to reduce their energy compared to high income occupants because low-income occupants are more likely to reside in older buildings with poor envelope conditions. Many studies found that income has a significant positive effect on residential energy consumption; that is, the higher the residents' income level, the greater their energy consumption [18,24,26,32,41,56,58,59]. [59] determined that in Ireland, occupants whose annual incomes is more than 30,000 Irish pounds (~$41,000) consumed more energy than those occupants with low annual income of 10,000 Irish pounds (~$13,600). However, other studies reported that the income level of occupants is complicated variable in analyzing its relationship with occupants' energy consumption [61]. On the other hand [31], found that there is a mixed relationship between occupants' income and their energy consumption behavior.

Employment Status. [46] stated that higher professionals consume more energy than lower professionals [62]. found that retired

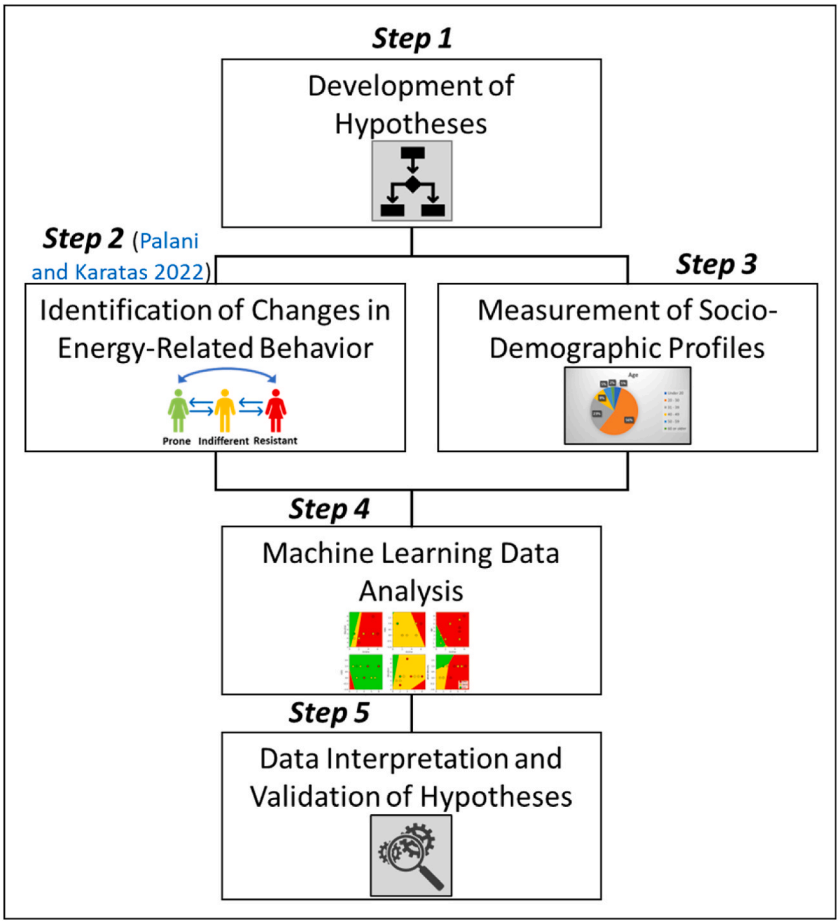


Fig. 1. Understanding the interrelationship between socio-demographics and energy-use profile of occupants.

occupants consume significantly less energy than employed occupants [27]. indicated a similar effect in that retired occupants consume less energy than employed occupants but not statistically significant. Yet, several studies found that employment status does not has an impact on the energy consumption of occupants [31,41].

2.2. Bibliometric analysis

A search of relevant articles was run using the Scopus database. Since this study focuses on identifying the relationship between energy-related behavior of occupants and their socio-demographic variables, the search subject area focused on related keywords such as "Energy Consumption Behavior," "Socio-demographics," "Buildings," "Residential," "Dormitories," "Offices," and "Hotels." These keywords were included in the initial search under the "title/abstract/keyword" section of Scopus. In Scopus database search, the quotation marks (" ") indicate an approximate or loose phrase and use of an asterisk (*) as a wildcard to ensure you capture more results. In this search, only those papers were filtered that were written in English language and was no limitation on the year that the paper was published. First, the following keywords were searched ("Energy Use* Behavior*" OR "Energy Consum* Behavior*" OR "Electric* Use* Behavior*" OR "Electric* Consum* Behavior*"). Then, a new search field was added with the keyword ("Building*"). The total number of related articles identified was 279. Next, we added another search field to categorize the building types. The keywords and results were classified as follows: (1) residential: adding the keywords ("Residential*" OR "Home" OR "House" OR "Household" OR "Neighborhood"), we found that there are 149 articles related to occupants' energy consumption behavior in residential building (57%; 159 out of 279 articles); (2) offices: adding the keyword ("office*"), we found that there are 51 related articles (18%); (3) dormitories: adding the keywords ("Dorm*" OR "University*" OR "Campus*"), we found 39 related articles (14%); (4) hotels: adding the keyword ("Hotel*"), we found only 3 related articles (1%); and (5) related articles in other building types is 27 articles (10%). After that, we added a new search field with the keywords ("Socio-demographic*" OR "Demographic*). As a result, there are 57 articles remained that investigated the interrelationships between energy consumption behavior of occupants and their socio-demographic variables, and 48 articles out of these 57 articles were only conducted on residential buildings. These studies provide valuable insight into the relationship between socio-demographic profiles of occupants and their energy consumption behavior in residential buildings. However, inconsistent findings in previous studies are due in part to variations in residential building characteristics (e.g., layout, insulation type) available to occupants. In contrast, hotel buildings provide a more uniform structure and level of accessibility to features, allowing for accurate measurement of energy-use behavior and identification of its relationship with socio-demographic and economic variables.

3. Methodology

The methodology for this study was developed and implemented in five steps (see Fig. 1): (1) develop a set of hypotheses to examine the relationship between socio-demographic and economic characteristics of occupants and their energy-related behavior (i.e., energy-use behavior and energy-use profiles); (2) identify changes in occupants' energy related behavior before and after being exposed to interventions by conducting a longitudinal study of two survey; (3) identify and record the socio-demographics of the occupants; (4) analyze and investigate various machine learning algorithms to measure the relationship between energy-related behavior of occupants and their socio-demographic and economic profiles; and (5) test the study hypotheses. Each step is explained in detail below.

Step 1 develops six hypotheses to investigate and identify the relationship between socio-demographic and economic characteristics of occupants and their energy-related behavior. The six hypotheses of this study are as follows.

- **Hypothesis 1 (H1):** There is a mixed relationship between occupants' energy saving behaviors and their age.
- **Hypothesis 2 (H2):** There is a mixed relationship between occupants' energy saving behaviors and their education level.
- **Hypothesis 3 (H3):** There is a mixed relationship between occupants' energy saving behaviors and their income.
- **Hypothesis 4 (H4):** Occupants' energy saving behaviors vary based on their gender.
- **Hypothesis 5 (H5):** Occupants' energy saving behaviors vary based on their race/ethnicity.
- **Hypothesis 6 (H6):** Occupants' energy saving behavior vary based on their employment status.

Step 2 identifies changes in occupants' energy-related behavior resulting from interventions; a longitudinal study was conducted using two surveys – pre-intervention and post-intervention surveys– each consisting of 52 questions. The pre-intervention survey

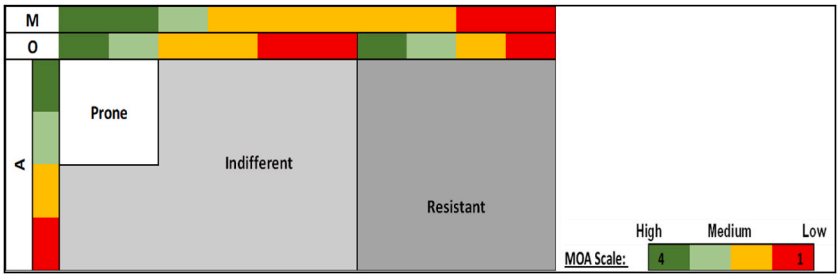


Fig. 2. Identification of energy-use profiles matrix [10,13].

collected data on (1) measuring Motivation-Opportunity-Ability (MOA) level of occupants to determine their energy-related behavior and to assess their readiness for change, and (2) measuring socio-demographic and economic characteristics such as age, gender, education level, ethnic background, income, and employment status (step 3 of the methodology section). Motivation (M) captured the occupants' inclination and readiness to adopt energy-saving practices. Opportunity (O) represented the environmental factors surrounding the occupants that are not under their control to take the desired actions. Ability (A) captured the occupants' prior knowledge and understanding of energy-saving practices, including their awareness of the associated benefits and conservation strategies [13]. K-means clustering was used to categorize occupants' energy-use behavior, which enabled the authors to identify energy-use behavior at different levels of MOA prior to proposing interventions and incentives. Additionally, an energy-use profile matrix was used to classify occupants as prone to change, indifferent to change, or resistant to change. For example, an occupant with Low Motivation (color-coded in red), Medium-Low Opportunity (color-coded in orange), and High Ability energy-use behavior (color-coded in green) would have a Resistant to Change energy-use profile, as determined by the energy-use profile matrix (Fig. 2). This is thoroughly explained and implemented in Ref. [13]. The term "resistant to change" refers to occupants who are unwilling to adopt energy-saving behavior, while "indifferent to change" describes those who are neutral in their willingness to adopt such behavior. On the other hand, "prone to change" describes occupants who are eager to adopt energy-saving behavior.

After categorizing occupants into energy-use profile groups, the proposed interventions and incentives were assigned to each group based on their level of readiness for change, as illustrated in Fig. 3. These interventions and incentives were developed in collaboration with a team of hotel owners who have a chain of over 60 hotels in the United States.

The second survey (i.e., post-intervention survey) was conducted with the same participants from the first survey to measure the changes in occupants' energy-related behavior after receiving the proposed interventions and incentives. The survey used the Motivation-Opportunity-Ability (MOA) framework to measure occupants' level of motivation, opportunity, and ability in adopting energy-saving practices. This allowed the researchers to identify changes in energy-related behavior and energy-use profiles of occupants after receiving the interventions.

The K-means clustering algorithm was used again to classify occupants' energy-use behavior after the interventions, which enabled them to determine the impact of the interventions on different MOA levels. The occupants were clustered into four energy-use behavior groups (Low, Medium-Low, Medium-High, and High) for each MOA level and categorized into three energy-use profile groups (resistant to change, indifferent to change, and prone to change) based on the energy-use profile matrix.

Finally, the changes in occupants' energy-related behavior before and after receiving the interventions were identified. This is explained in detail in Ref. [14].

In this study, it was assumed that each occupant's MOA level has four energy-use behaviors and three energy-use profiles based on previous findings. Thus, the four energy-use behaviors were converted to numeric data with "1" representing Low energy-use behavior, "2" representing Medium-Low, "3" representing Medium-High, and "4" representing High energy-use behavior. Likewise, the three energy-use profiles were also converted to numeric data, with "1" indicating Resistant to Change energy-use profile, "2" representing Indifferent to Change energy-use profile, and "3" representing Prone to Change energy-use profile.

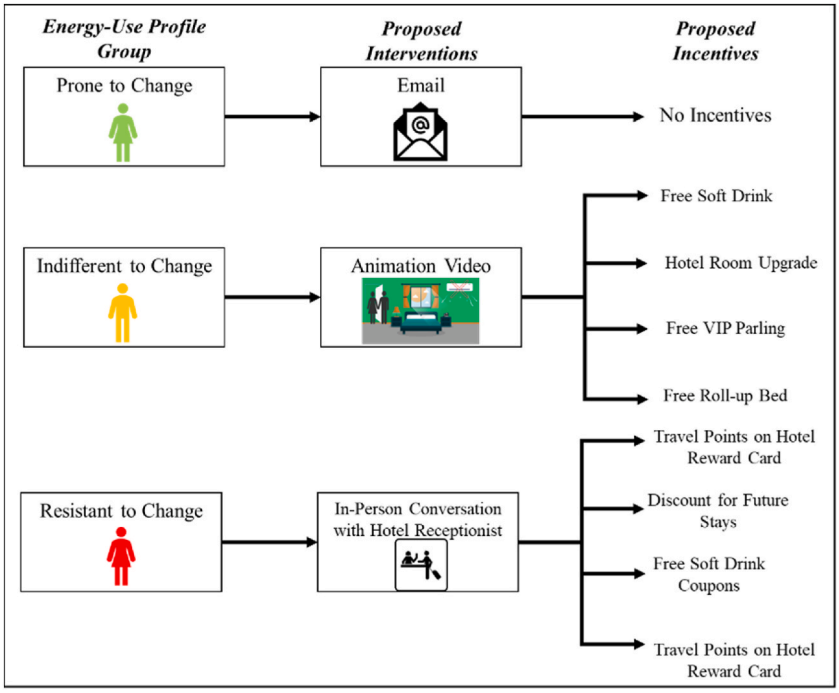


Fig. 3. Proposed Interventions and Incentives to Energy-Use Profile Groups; deployed from [14].

Step 3 measures and records the socio-demographic and economic variables of the occupants who participated in the longitudinal study of two surveys. The socio-demographic and economic variables in this study include age, gender, education level, ethnic background, income, and employment status.

Step 4 investigates various machine learning algorithms to visualize, measure, and interpret the relationship between socio-demographic profiles of occupants and changes in their energy-related behavior before and after being exposed to the proposed education-based interventions and incentives shown in Fig. 3. Specifically, we implemented linear regression, ordered logistic modelling, decision tree learning, and support vector machines algorithms to find the best fit and to capture complex relationships.

Step 5 tests the six proposed hypotheses listed above. This step was achieved with the machine learning algorithms from step 4.

4. Data

The data for this study was collected through a survey conducted on Prolific, an online platform for recruiting participants for surveys. This method has been used in similar studies, such as those by Anderson et al. (2017) and [63]; to measure energy use behavior and provide feedback to occupants. The advantage of this methodology is that it allows the authors to capture a diverse range of energy-related behaviors from hotel guests in multiple hotels building who stayed at various types of hotels in the United States, including three-, four-, and five-star hotels instead of collecting data in a single hotel building. In the pre-intervention survey, 405

Table 2
Selected socio-demographic variables, their values, and number of samples.

Characteristics	Socio-demographic variables	Values/Categories	# Of Samples Before Interventions (n = 370)	# Of Samples After Interventions (n = 211)
Ordered	Age	1. Under 20	1.5% (n = 19)	1.4% (n = 8)
		2.20–30	2.56% (n = 207)	2.58% (n = 123)
		3.31–39	3.23% (n = 85)	3.21% (n = 44)
		4.40–49	4.9% (n = 33)	4.10% (n = 21)
		5.50–59	5.5% (n = 19)	5.4% (n = 8)
		6.60 and older	6.2% (n = 7)	6.3% (n = 6)
		7. Prefer not to answer	7.0% (n = 0)	7.0% (n = 0)
	Education Level	1. No schooling completed, or less than 1 year	1.0% (n = 0)	1.0% (n = 0)
		2. Nursery, kindergarten, and elementary (grades 1–8)	2.0% (n = 0)	2.0% (n = 0)
		3. High school (grades 9–12, no degree)	3.2% (n = 7)	3.2% (n = 4)
		4. High School graduate (or equivalent)	4.30% (n = 111)	4.30% (n = 64)
		5. Bachelor's degree (BA, BS, AB, etc.)	5.40% (n = 148)	5.39% (n = 82)
		6. Master's degree (MA, MS, MENG, MSW, etc.)	6.17% (n = 63)	6.18% (n = 38)
		7. Professional degree (MD, DDC, JD, etc.)	7.1% (n = 4)	7.2% (n = 4)
		8. Doctorate degree (PhD, EdD, etc.)	8.3% (n = 11)	8.4% (n = 8)
		9. Associate degree (AA, AS, AAA, AAS)	9.7% (n = 26)	9.5% (n = 11)
	Income	1. Less than $20,000	1.22% (n = 81)	1.21% (n = 45)
		2. $20,000 to $34,999	2.18% (n = 66)	2.18% (n = 38)
		3. $35,000 to $49,999	3.14.5%(n = 54)	3.16% (n = 33)
		4. $50,000 to $74,999	4.18% (n = 66)	4.18% (n = 38)
		5. $75,000 to $99,999	5.11.5%(n = 43)	5.12% (n = 26)
		6. Over $100,000	6.9% (n = 34)	6.9% (n = 18)
		7. Prefer not to answer	7.7% (n = 26)	7.6% (n = 13)
Nominal	Gender	1. Male	1.52% (n = 192)	1.51% (n = 108)
		2. Female	2.48% (n = 178)	2.49% (n = 103)
		3. Prefer not to answer	3.0% (n = 0)	3.0% (n = 0)
	Race/Ethnicity	1. White	1.57% (n = 211)	1.60% (n = 126)
		2. Hispanic, Latino, or Spanish Origin	2.11% (n = 41)	2.10% (n = 21)
		3. Black or African American	3.6% (n = 22)	3.5% (n = 11)
		4. Asian	4.17% (n = 63)	4.15% (n = 32)
		5. American Indian or Alaska Native	5.0% (n = 0)	5.0% (n = 0)
		6. Middle Eastern or North African	6.5% (n = 18)	6.6% (n = 13)
		7. Native Hawaiian or Other Pacific Islander	7.0% (n = 0)	7.0% (n = 0)
		8. Other	8.3% (n = 11)	8.3% (n = 6)
		9. Prefer not to answer	9.1% (n = 4)	9.1% (n = 2)
	Employment Status	1. Employed full time (40 or more hours per week)	1.43% (n = 159)	1.41% (n = 86)
		2. Employed part time (up to 39 h per week)	2.23% (n = 83)	2.23% (n = 48)
		3. Unemployed	3.10% (n = 38)	3.9% (n = 20)
		4. Student	4.21% (n = 78)	4.23% (n = 48)
		5. Retired	5.1% (n = 4)	5.1% (n = 3)
		6. Prefer not to answer	6.2% (n = 8)	6.3% (n = 6)

responses were collected. After data clean-up and the removal of inconsistent responses, the total number of responses decreased to 370. For the post-intervention survey, all the 370 respondents were invited to participate. A total of 270 responses were collected out of the 370 participants invited. After data clean-up, the total number has been reduced to 211 responses (i.e., 57% response rate). Accordingly, 211 valid responses were considered for the data analyses.

The Scikit-learn library in Python as well as the Pandas library were used to implement machine learning algorithms [64–66]. The details of the six socio-demographic questions in the surveys along with values and number of samples in both surveys are shown in Table 2.

Given that all the variables were categorical, those that were not ordered were hot encoded into binary columns with 0 or 1 for each possible category. The ordered variables were left untouched so the model can capture the meaning of having higher/lower levels of income or education as well as belonging to a younger or lower population.

For modelling purposes, the dataset was divided into two: 80% for training and 20% for testing. In addition, given that the dataset was imbalanced towards one of the classes, the Synthetic Minority Oversampling Technique (SMOTE) was implemented to oversample the samples in the minority classes only in the training dataset. Further, a 5-fold cross-validation procedure was implemented on the train dataset. Namely, the training dataset was further divided into five subsets of data, four were used for training and one for testing, and the operation was repeated five times so that each subset is used for testing once. Finally, the results are averaged to determine the overall accuracy of the model. This 5-fold cross validation procedure is also used to select the model hyperparameters. For linear regression, the model hyperparameters are just the slope and intercept. For Decision Tree, they are the split criterion, maximum depth, minimum samples, and maximum variables. The decision tree, random forest, and gradient boosting models hyperparameters were tuned using a grid search algorithm. The hyperparameters of the three tree-based models and the SVM estimated are summarized in Table 3.

5. Results on predicting the relationship between socio-demographics and energy-use behavior

5.1. Model performance

A categorical variable was created for each MOA specifying whether the value increased, decreased, or stayed the same. Six different models were calibrated for each MOA level: Linear Regression (LR), Ordered Logistic Regression (OLR), Support vector machines (SVM), Decision Tree (DT), Random Forest (RF), and Gradient Boosting (GB). Each model makes use of the demographic information available. For each machine learning method, the 5-fold cross-validation process was first implemented and then the model was tested on the test data (see Table 4). The machine learning models were built to predict one of three classes: 'Decr', 'Same', and 'Incr', which indicated whether the corresponding MOA value decreased, stayed the same, or increased respectively.

Table 4 illustrates that Motivation is the target variable with the highest train and test accuracies overall. It can also be observed that DT, GB, and SVM perform well for our dataset. In the end, we selected DT as the preferred method despite its lower accuracy on the test set because the tree learned is easier to interpret (knowing that the goal of this article is to interpret the results not to achieve the highest modeling accuracy).

Moreover, the lowest levels of accuracy on both the test and train sets achieved for the Ability and Opportunity variables make them unsuitable for deriving equivalent observations as the ones with the Motivation. These results indicate that these two levels of Motivation-Opportunity-Ability (MOA) are not effective in determining the interrelationship between the changes in occupants' energy-use behavior and their socio-demographic profiles. These findings can be justified based on the definition of both Opportunity and Ability. Opportunity refers to the surrounding environmental factors that are not under the control of occupants to enable desired energy saving actions, which is an external factor not related with occupants' socio-demographic profiles. Ability refers to occupants' prior knowledge about possible energy conservation strategies, which is also not considered to have a direct relationship with occupants' socio-demographic profiles. Hence, among the MOA variables, the Motivation is the most prone to be modelled with either DT or SVM machine learning methods.

The results from the six applied machine learning methods are presented in the form of a confusion matrix, which contains the predicted classes as columns and the real classes as rows (see Table 5). The rows represent the ground truth while the columns represent the predicted observations. For instance, in Table 5, the element in row 1 (Improve) and column 2 (Same) of the Ordered Logit confusion matrix means that there were 12 observations that should have been classified as having improved their Motivation, but all 12 were mistakenly classified as keeping the same Motivation. In the ideal scenario in which all observations are properly classified, all the matrix elements would be zero except for the diagonal. Therefore, the Ordered Logit regression fails to properly capture the observations that lead to a change in MOA attributes. This behavior was observed also for Ability and Opportunity. From the confusion matrix, we can see that while DT does capture complex relationships as GB and RF, because we are dealing with deterministic system and with few data points, DT is suitable to interpret our results.

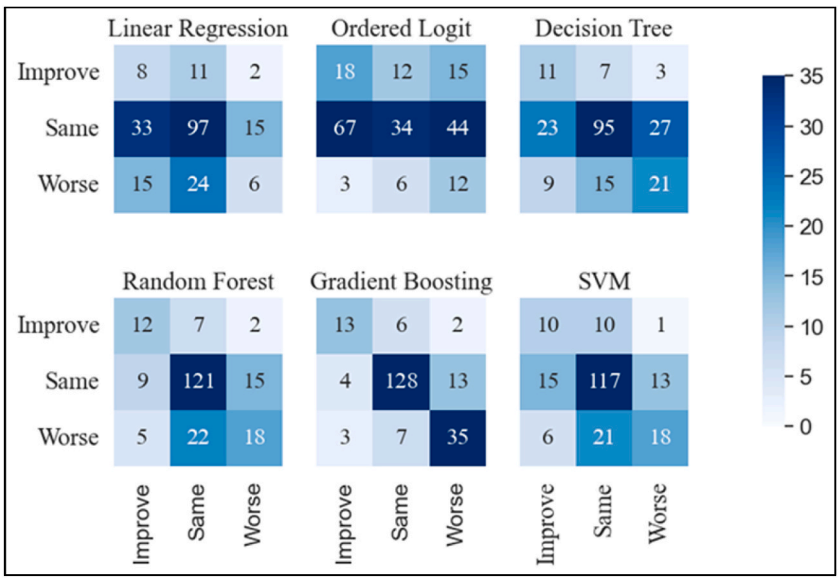
Table 3
Hyperparameters of the tree-based and SVM motivation models estimated.

Machine Learning Methods	Criterion	Maximum Depth	Number of Estimators	Learning Rate
Decision Tree	Entropy	5	–	–
Random Forest	Entropy	5	100	–
Gradient Boosting	Friedman mse	5	500	0.1
	C		**Kernel**	
SVM	1000	poly		

Table 4
Machine learning algorithms accuracies for modelling changes (increased, decreased or stayed the same) in MOA.

	Motivation		Opportunity		Ability	
	Train	Test	Train	Test	Train	Test
Linear Regression (LR)	52.0%	46.9%	48.4%	48.4%	44.3%	20.3%
Ordered Logistic Regression (OLR)	43.9%	20.9%	43.4%	30.2%	48.7%	39.5%
Decision Tree (DT)	71.6%	42.2%	67.5%	37.5%	70.2%	40.6%
Random Forest (RF)	75.6%	59.4%	93.3%	46.9%	86.6%	54.7%
Gradient Boosting (GB)	93.4%	60.0%	93.3%	50.0%	93.5%	54.7%
Support Vector Machines (SVM)	71.3%	57.8%	60.7%	43.8%	71.1%	46.9%

Table 5
Confusion Matrix for Predicting Motivation for Entire Dataset under Six Different Models.



5.2. Model interpretation

One additional insight is the feature importance (FI) (see Fig. 4). This metric measures which variables have a greater effect on the output of the model. In order to interpret our model, the DT algorithm was applied (see Fig. 4). The FI plot suggests that age is the most important variable to predict the energy consumption behavior of occupants when they exposed to education-based interventions and incentives. In addition, income, education, and gender have high importance too. For the sake of result consistency, we applied all the three tree-based algorithms (only tree-based algorithms have FI results as shown in the Supplementary Materials. Since the SVM was

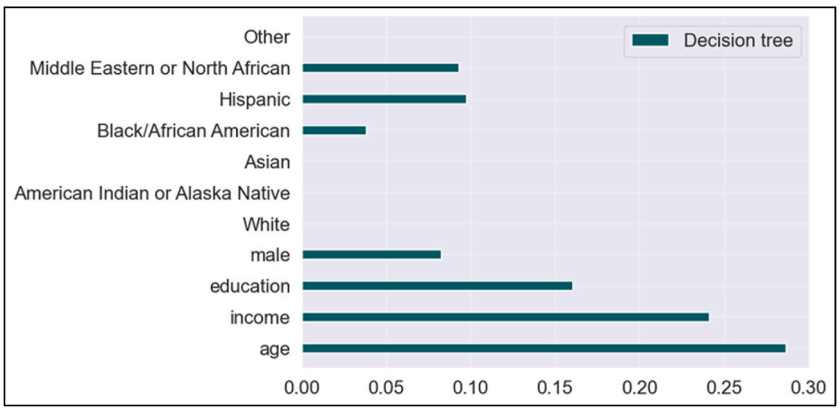


Fig. 4. Feature importance for decision tree method methods for the prediction of occupants' motivation level change.

fitted by means of a transformation in another vector space, it is not possible to offer such feature importance insight for that model, which is ultimately a reason to choose DT based models over SVM for the sake of interpretability.

6. Results on predicting the interrelationship between socio-demographics and energy-use profile

The energy-use profiles and changes in energy-use profiles of the 211 participants were identified before and after being exposed to the proposed interventions. Fig. 5 shows the occupants energy-use profiles before and after the interventions as well as the percentage changes in their energy-use profiles. The findings demonstrate that there was a 52% increase in the Prone to Change energy-use profile, a 3% increase in the Indifferent to Change energy-use profile, and a 23% decrease in the Resistant to Change energy-use profile. This is a promising and valid indication that the proposed interventions were effective in encouraging occupants to change their energy consumption behavior in hotel buildings for the better.

6.1. Model performance

This study also investigated the possibility of predicting the relationship between socio-demographics of occupants and their energy-use profile (i.e., pre-, and post-intervention surveys' participants). For this task, we proceeded in an analogous way as in the energy-use behavior case by modelling the change in energy-use profiles to model whether the energy-use profiles of occupants got improvement, remained the same or got worse before and after the occupants being exposed to the proposed interventions. Four machine learning models were fitted in two different scenarios: considering the Motivation before proposing the interventions as a variable and without considering it (see Table 6).

The first important thing to notice from the results in Table 6 is that there is no significant difference between including Motivation as a variable or not. Including the Motivation results even in a slight decrease in accuracy in some cases. In addition, in this case DT and GB have all similar performances, but again for the convenience of visualization and interpretability, DT is preferred.

6.2. Model interpretation

Similar to the process applied with the energy-use behavior prediction section, DT was applied to interpret the results and predict the interrelationship between socio-demographic variables and energy-use behavior using FI diagram. The corresponding variable importance for energy-use profiles can be found in Fig. 6. This plot shows that the proposed models in this study rely on fewer variables. Again, to validate the accuracy of our results, we also applied RF and GB to show all the three tree-based algorithms (see the Supplementary Materials).

In this section, we also visualized the DT model by estimating energy-use profile changes based on socio-demographic variables. The trained tree is shown in the Supplementary Materials. This diagram offers a schematic view of the rules learned by the tree and the metrics at each node. The entropy metric refers to the impurity of the node, which represents the probability for a randomly chosen element to be misclassified. The color of the nodes represents the majority class for classification (purple: worse, orange: improves, green: stayed the same). A stronger color shade indicates a higher purity given by entropy. The sample value represents the number of samples at each node and the value vector is the number of samples per class. Finally, the class is the label of the observed socio-demographics at the node if there is no further splitting. To make it easier to understand, we presented the DT analysis in a pseudo-tree (see Fig. 7). From the DT, four different segments show occupants who are likely to switch to better energy-use profiles after being exposed to the proposed interventions, while one segment indicates occupants who are expected to change their energy-use profiles to worse. The findings of the decision tree are summarized in Table 7.

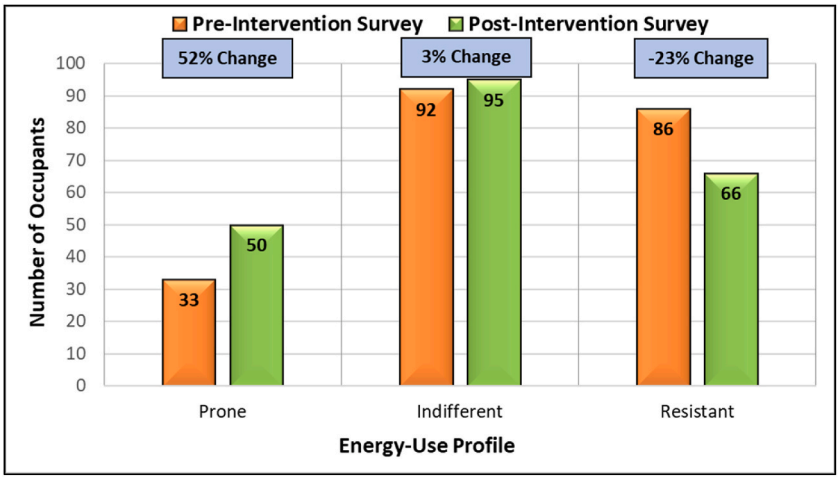


Fig. 5. Changes in occupants' energy-use profile before and after exposed to the proposed interventions [14].

Table 6
Machine Learning Models Accuracies for Modelling Changes (increased, decreased or stayed the same) in Energy-Use Profiles with and without Motivation level.

	Energy Profile (without Motivation)		Energy Profile (with Motivation)	
	Train	Test	Train	Test
Decision Tree (DT)	73.0%	51.2%	72.3%	50.0%
Random Forest (RF)	82.1%	46.9%	92.2%	42.2%
Gradient Boosting (GB)	82.1%	56.2%	93.0%	46.9%
Support vector machines (SVM)	64.6%	20.3%	71.7%	26.6%

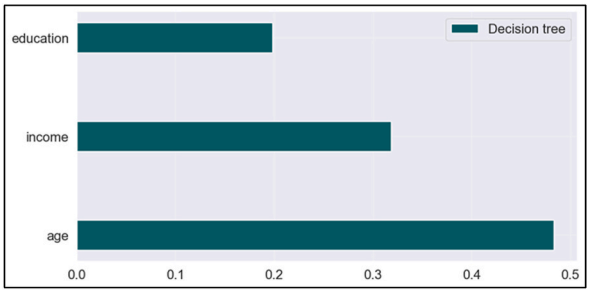


Fig. 6. Feature importance for DT method for the prediction of energy-use profiles.

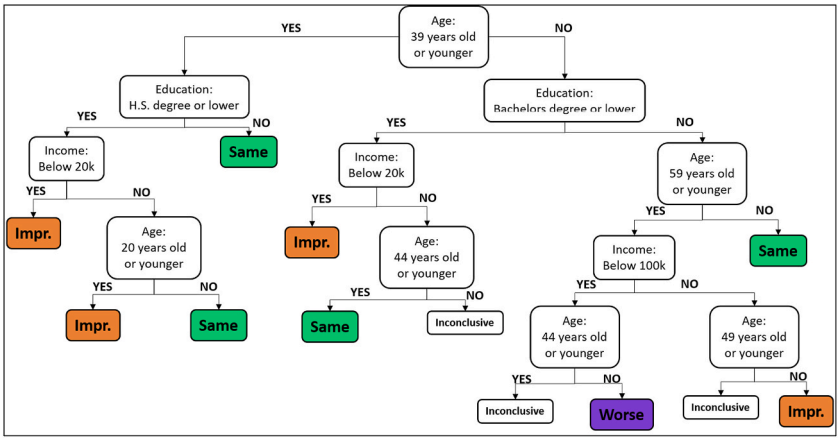


Fig. 7. Pseudo-tree of the model estimating energy-use profile changes from demographic variables.

7. Testing the hypotheses

The six proposed hypotheses were tested and validated based on the machine learning algorithms used in this study. The findings are summarized in Table 8 and explained in detail below.

Hypothesis 1. **(H1):** the tree in Fig. 6 shows that all classes at the bottom of the tree involve a node with an age-based decision. Moreover, the feature importance plot in Fig. 5 shows age as the feature with the highest importance. These results suggest that there is mixed relationship between occupants' energy saving behaviors and their age. Therefore, H1 was accepted. This finding is consistent with previous studies [18,26,29,31,37,47,59,67]. It was also found that there is a relationship between occupants' energy saving behaviors and their age if considering income and education as additional variables (see Table 7).

Hypothesis 2. **(H2):** the tree in Fig. 6 shows that the education level is present at the top of all ramifications of the tree. The education level is also the second most important feature from Fig. 5. These results suggest that there is mixed relationship between occupants' energy saving behaviors and their education level. Therefore, H2 was accepted. This finding is consistent with previous studies [29,40,57]. It was also found that there is a relationship between occupants' energy saving behaviors and their education level if considering income and age as additional variables (see Table 7).

Hypothesis 3. **(H3):** the tree in Fig. 6 shows that most classes at the bottom of the tree involve a node with an income-based decision. In addition, the feature importance plot in Fig. 5 shows income as the third feature with the highest importance. This result shows that there is mixed relationship between occupants' energy saving behaviors and their income. Therefore, H3 was accepted. This finding is

Table 7
Segments Showing the Occupants that are Prone to Improve or Worse their Energy-Use Profiles.

Energy-Use Profile Improve/Worse	Segment No.	Segment Description	Energy-Use Profiles Before Interventions	Energy-Use Profiles After Interventions	Provided Interventions
Improve	1	- Below 40 years old - Highschool education or lower - Income below 20 k	Indifferent to Change	Prone to Change	Animation Video
	2	- 20 years old or younger - Highschool education or lower - Income above 20 k	Resistant to Change	Indifferent to Change	In-Person Conversation with Hotel Receptionist
	3	- 40 years old or older - Bachelor's degree or lower - Income below 20 k	Resistant to Change	Indifferent to Change	In-Person Conversation with Hotel Receptionist
	4	- Master's degree or higher - Between 50 and 59 years old - Income above 100 k	Indifferent to Change	Prone to Change	Animation Video
Worse	5	- Master's degree or higher - Between 45 and 59 years old - Income below 100 k	Prone to Change	Indifferent to Change	Email

Table 8
Accepting/Rejecting the Hypotheses and Identify the Type of Relationship between Occupants' Energy Saving Behaviors and their Socio-demographic and Socio-economic Characteristics.

Hypothesis No.	Hypothesis	Accept/Reject Hypothesis	Type of Relationship
H1	There is a mixed relationship between occupants' energy saving behaviors and their age.	Accepted	Mixed Relationship
H2	There is a mixed relationship between occupants' energy saving behaviors and their education level.	Accepted	Mixed Relationship
H3	There is a mixed relationship between occupants' energy saving behaviors and their income.	Accepted	Mixed Relationship
H4	Occupants' energy saving behaviors vary based on their gender.	Rejected	No Relationship
H5	Occupants' energy saving behaviors vary based on their race/ethnicity.	Rejected	No Relationship
H6	Occupants' energy saving behavior vary based on their employment status.	Rejected	No Relationship

consistent with previous studies [61,68]. It was also found that there is a relationship between occupants' energy saving behaviors and their income if considering age and education level as additional variables (see Table 7).

Hypothesis 4. **(H4):** gender was not found to be a relevant feature from the feature importance plot in Fig. 5. Therefore, H4 was rejected. This finding is consistent with previous studies [52,53].

Hypothesis 5. **(H5)**: none of the race and ethnicity categories were found to be relevant in Fig. 5. Therefore, H5 was rejected. This finding is consistent with previous studies [39].

Hypothesis 6. **(H6)**: none of the employment status classes were found to have a significant impact in the model according to the feature importance plot in Fig. 5. This result illustrates that there is no relationship between occupants' energy saving behaviors and their employment status. Therefore, H6 was rejected. This finding is consistent with previous studies [31,41].

8. Research limitations

This study has some limitations, which can be summarized as follows. Firstly, the decision-based methods used in the study have superior predictive performance in real-world scenarios compared to more traditional techniques like linear and logistic regressions. However, these methods are prone to overfitting, especially when there are multiple levels involved. To overcome this issue, the authors restricted the depth of all trees. Secondly, trees can be biased towards the dominant class, which was the "Same" class in this study as shown in Table 5. To mitigate this, the authors tried to balance the data as much as possible. Nonetheless, the decision tree confusion matrix in Table 5 suggests that the tree can still make predictions for other classes, indicating that this approach may benefit from a larger sample size and proper balancing in future applications. Finally, variations in input data can produce different trees, which introduces an element of instability.

9. Conclusion

In this study, six models (e.g., linear regression, ordered logistic models, decision tree learning, random forest, gradient boosting, and support vector machines) were implemented to examine and understand the interrelationship between the energy consumption behavior of occupants and their socio-demographic and economic characteristics (i.e., age, gender, education level, race/ethnic background, income, and employment status) before and after being exposed to proposed interventions (i.e., email, animation video, and in-person conversation with hotel receptionist) and incentives (e.g., free soft drink, free VIP parking, travel points on hotel reward card, discount for future stay, etc.).

The decision-based methods used in this study have demonstrated superior performance in predicting test data with similar levels of accuracy. While the decision tree (DT) algorithm may have lower accuracy compared to GB as indicated in Tables 6 and it provides higher interpretability by generating a single tree with decision rules that can be examined directly. This enables the readers to gain full insights into the process within the tree. Given that many machine learning (ML) methods have been criticized for being black boxes, the interpretability feature of the DT method is highly valuable. Therefore, the DT method has been chosen as the most suitable approach for this study.

Additionally, the DT analysis of the data has revealed significant insights into the occupants' response to the proposed interventions. The findings demonstrate the presence of five distinct segments, each representing a specific group of occupants with unique socio-demographic and economic characteristics. Interestingly, these segments showed a strong potential for improving their energy-use profiles after being exposed to the interventions. However, one segment displayed a contrary outcome, experiencing a decline in their energy-use profiles. A comprehensive summary of these results can be found in Table 7. Notably, it was observed that the four segments, namely Segment 1 to Segment 4, which demonstrated a likelihood of enhancing their energy-use profiles after interventions, were initially categorized as Indifferent or Resistant to Change. Segment 1 and Segment 4, initially categorized as Indifferent to Change, displayed notable improvements in their energy-use profiles after receiving the animation video intervention. Similarly, Segment 2 and Segment 4, initially categorized as Resistant to Change, exhibited enhanced energy-use profiles following in-person conversation with hotel receptionist intervention. These findings highlight the effectiveness of the proposed interventions, specifically the animation video and in-person conversation interventions, in successfully influencing occupants' energy-use profiles for the better. However, the email intervention provided to the occupants that are Prone to Change yielded less favorable results compared to the other two interventions. In fact, Segment 5, consisting of occupants who were Prone to Change prior to the interventions and then received the email intervention, experienced a worsening of their energy-use profiles after the email intervention.

The findings can help decision-makers such as utility companies and policy makers propose effective and sufficient strategies and education-based interventions such as video animation and in-person conversation with hotel receptionist intervention based on the socio-demographics of occupants to minimize energy consumption in buildings. It may be easier for decision makers to scale up the findings of this study to a larger scale such cities, neighborhoods, and towns. This will help categorize occupants into different energy consumption behavior groups based on occupants' socio-demographic and economic characteristics and their energy-use profiles to better control and monitor building energy consumption. Accordingly, this in turn may help energy providers to propose interventions and incentives to occupants in those cities, neighborhoods, or towns that are Resistant or Indifferent to Change their energy-use profiles and provide interventions with some kind of incentives to those occupants that mostly are Prone to Change to keep their good energy consumption behavior.

Author contributions

Hevar Palani and Aslihan Karatas: Conceptualization, Methodology, Validation, Resources, Data Curation, Visualization. **Hevar Palani:** Investigation, Writing—Original Draft Preparation. **Juan G. Acosta and Hevar Palani:** Software, Formal Analysis. **Aslihan Karatas and Sybil Derrible:** Writing—Review & Editing, Supervision, Project Administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jobe.2023.106875.

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